

Vision Tokens and Multimodal Alignment

Mathematics of Deep Learning — Chapter 15

Yin Yang

Foundation Books Project

Roadmap

From Images to Patch Tokens

Patch Embeddings, Class Tokens, Positions

Vision Transformer Algebra

Pooling and Image Representations

Shared Embedding Geometry

Contrastive Image-Text Learning

Zero-Shot Classification

Retrieval and Ranking

Fusion and Cross-Attention

Diagnostics, Biases, Deployment

Summary

From Images to Patch Tokens

The Campus Image-Search Problem

A campus search system must put pixels and words into comparable coordinates.

- Running batch: matched image-caption pairs $(x_1, t_1), \dots, (x_4, t_4)$.
- x_1 : bicycle near building $\leftrightarrow$ “a bicycle parked near a building.”
- Two applied questions:
 - given text, which image is nearest?
 - given an image, which prompt names it best?
- Positive pair: matched (x_i, t_i) . Mismatched pair: (x_i, t_j) with $j \neq i$.

Matched pairs define the alignment target; mismatched pairs become contrastive negatives.

Image as a Tensor, Then a Patch Grid

Channel-last convention:

$$X \in \mathbb{R}^{H \times W \times C}.$$

For a patch size P with $P \mid H$ and $P \mid W$, non-overlapping patches produce

$$N = \frac{H}{P} \cdot \frac{W}{P}$$

patch tokens, each a flat vector in $\mathbb{R}^{P^2 C}$.

Image size	P	Tokens N	All-pair scores N^2
224×224	16	196	38416
384×384	16	576	331776
224×224	32	49	2401

Patch size trades spatial detail against the quadratic attention cost.

Patch Tokenization, Built Up

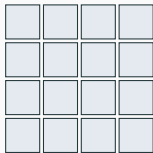
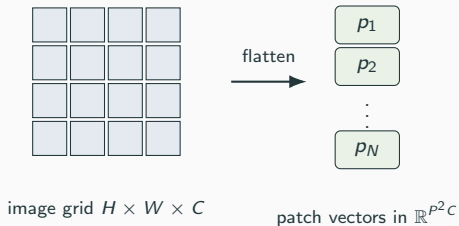


image grid $H \times W \times C$

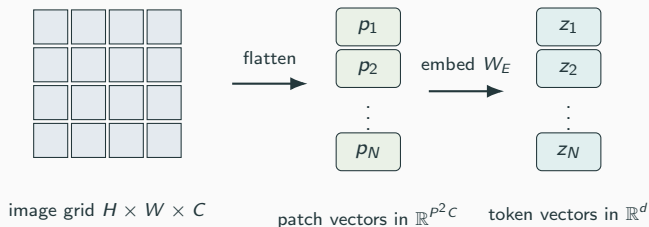
(1) Image is a tensor with spatial structure.

Patch Tokenization, Built Up



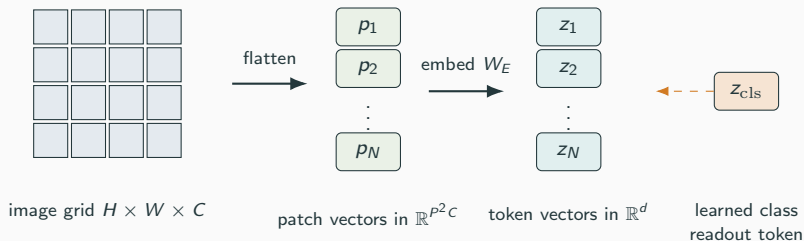
- (1) Image is a tensor with spatial structure.
- (2) Split into $P \times P \times C$ patches, flatten.

Patch Tokenization, Built Up



- (1) Image is a tensor with spatial structure.
- (2) Split into $P \times P \times C$ patches, flatten.
- (3) Affine map $e(p) = pW_E + b_E$ gives $\mathbb{R}^d$ tokens.

Patch Tokenization, Built Up



- (1) Image is a tensor with spatial structure.
- (2) Split into $P \times P \times C$ patches, flatten.
- (3) Affine map $e(p) = pW_E + b_E$ gives $\mathbb{R}^d$ tokens.
- (4) A class token is prepended as a learned readout.

Tokenization is deterministic; the embedding is trained.

Patch Embeddings, Class Tokens, Positions

Patch Embedding: An Affine Map

For one patch $p \in \mathbb{R}^{P^2 C}$,

$$e(p) = pW_E + b_E, \quad W_E \in \mathbb{R}^{P^2 C \times d}, \quad b_E \in \mathbb{R}^d.$$

- Parameter count: $P^2 Cd + d$.
- Example: $P = 16$, $C = 3$, $d = 768$ gives $768 \cdot 768 + 768 = 590,592$.
- Same affine layer applied independently to every patch.

Class token $z_{\text{cls}} \in \mathbb{R}^d$: a learned readout slot, not another image patch.

The patch embedding is the first trained parameter block a ViT sees.

Positions and the Initial Sequence

Positional embeddings $e_1^{\text{pos}}, \dots, e_T^{\text{pos}} \in \mathbb{R}^d$ carry grid information self-attention does not infer from content alone. Initial transformer input with $T = N + 1$ tokens:

$$Z_0 = [z_{\text{cls}}; z_1; \dots; z_N] + E_{\text{pos}}, \quad Z_0 \in \mathbb{R}^{(N+1) \times d}.$$

- Without position information, the patch grid is reduced to a bag of vectors.
- Class token and positions are added *before* the first attention block.
- Sequence length is $T = N + 1$; for $N = 196$, $T = 197$.

Position embeddings are how grid coordinates enter a permutation-equivariant operator.

Vision Transformer Algebra

Self-Attention on Visual Tokens

Let $Z \in \mathbb{R}^{T \times d}$. One head forms

$$Q = ZW_Q, \quad K = ZW_K, \quad V = ZW_V,$$

with $Q, K \in \mathbb{R}^{T \times d_k}$, $V \in \mathbb{R}^{T \times d_v}$. Then

$$S = \frac{QK^\top}{\sqrt{d_k}}, \quad A = \text{softmax}_{\text{row}}(S), \quad H = AV.$$

- Score matrix $S \in \mathbb{R}^{T \times T}$, entry $S_{i,j} = q_i^\top k_j / \sqrt{d_k}$.
- One score per ordered query-key pair: T^2 entries per head.
- Computing $QK^\top$ costs $O(T^2 d_k)$.

Attention cost grows with token pairs, not with pixel count directly.

Multi-Head Block and Parameter Count

With M heads and $d_k = d_v = d/M$ (no biases):

$$H^{(h)} = \text{softmax}_{\text{row}} \left(\frac{ZW_Q^{(h)} (ZW_K^{(h)})^\top}{\sqrt{d_k}} \right) ZW_V^{(h)}.$$

Concatenate heads, then apply $W_O \in \mathbb{R}^{d \times d}$.

- Q, K, V projections across all heads: $3Md(d/M) = 3d^2$.
- Output projection: d^2 .
- **Total** attention projection parameters: $4d^2$.
- Activation cost during training is proportional to MT^2 .

T^2 memory and $4d^2$ parameters are the two pressure points in a ViT block.

One Attention Row by Hand

Three value scalars 4, 1, -2 with attention weights (0.60, 0.25, 0.15):

$$0.60(4) + 0.25(1) + 0.15(-2) = 2.35.$$

- Weighted average of *values*, not of raw queries or keys.
- Softmax row sums to one; weights are nonnegative.
- Generalizes coordinate-wise to value vectors.

Attention maps are diagnostic views of pairwise token interaction, not complete explanations.

Attention is a weighted average whose weights are themselves learned from content.

Pooling and Image Representations

Pooling: From Token Sequence to One Vector

Classifier and retrieval heads need a single image vector $r(x) \in \mathbb{R}^d$. Class-token pooling:

$$r_{\text{cls}}(x) = z_0.$$

Average pooling over patch tokens:

$$r_{\text{avg}}(x) = \frac{1}{N} \sum_{i=1}^N z_i.$$

- Linear head: $o = r(x)W + b$, $W \in \mathbb{R}^{d \times K}$.
- Predicted class: $\arg \max_k o_k$.
- Choice of pooling interacts with pretraining and downstream task.

Pooling defines the single vector every downstream head sees.

Worked Pooling Example

Two patch tokens

$$z_1 = (2, 0), \quad z_2 = (0, 4).$$

Average pooling: $r_{\text{avg}} = (1, 2)$. Class vectors

$$w_{\text{bike}} = (2, 0), \quad w_{\text{cup}} = (0, 1), \quad w_{\text{bus}} = (-1, 1).$$

Logits:

$$\langle r, w_{\text{bike}} \rangle = 2, \quad \langle r, w_{\text{cup}} \rangle = 2, \quad \langle r, w_{\text{bus}} \rangle = 1.$$

Bike and cup tie; the decision rule must specify tie-breaking.

A representation is good *relative to* a task and a metric, not in isolation.

Shared Embedding Geometry

Two Encoders, One Vector Space

A dual-encoder vision-language model:

$$f_I : \mathcal{X}_I \rightarrow \mathbb{R}^d, \quad f_T : \mathcal{X}_T \rightarrow \mathbb{R}^d.$$

Comparable vectors require a shared embedding space. Normalize:

$$u_i = \frac{f_I(x_i)}{\|f_I(x_i)\|_2}, \quad v_j = \frac{f_T(t_j)}{\|f_T(t_j)\|_2}.$$

Image-text score:

$$s_{i,j} = u_i^\top v_j = \frac{f_I(x_i)^\top f_T(t_j)}{\|f_I(x_i)\|_2 \|f_T(t_j)\|_2}.$$

Normalization turns dot products into pure angle comparisons.

Similarity Matrix, Built Up

	t_1	t_2	t_3	t_4
x_1				
x_2				
x_3				
x_4				

(1) Rows = images, columns = captions.

Similarity Matrix, Built Up

	t_1	t_2	t_3	t_4
x_1	0.82	0.21	0.36	0.28
x_2	0.17	0.76	0.25	0.31
x_3	0.33	0.18	0.79	0.41
x_4	0.26	0.39	0.35	0.74

- (1) Rows = images, columns = captions.
- (2) Entry $s_{i,j}$ is normalized $u_i^\top v_j$.

Similarity Matrix, Built Up

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row x_3 : largest is 0.79 at t_3
 $\Rightarrow$ image x_3 retrieves caption t_3

- (1) Rows = images, columns = captions.
- (2) Entry $s_{i,j}$ is normalized $u_i^\top v_j$.
- (3) Matched pairs sit on the diagonal.
- (4) Retrieval = argmax along a row or column.

Cross-modal retrieval becomes a matrix-comparison problem.

Contrastive Image-Text Learning

Temperature-Scaled Logits and the CLIP Loss

Temperature $\tau > 0$:

$$a_{i,j} = \frac{u_i^\top v_j}{\tau}.$$

Image-to-text loss for image i :

$$\ell_i^{I \rightarrow T} = -\log \frac{\exp(a_{i,i})}{\sum_{j=1}^m \exp(a_{i,j})}.$$

Symmetric batch loss:

$$\mathcal{L}_{\text{sym}} = \frac{1}{2} \mathcal{L}_{I \rightarrow T} + \frac{1}{2} \mathcal{L}_{T \rightarrow I}.$$

Each row is a softmax classifier whose classes are the in-batch captions.

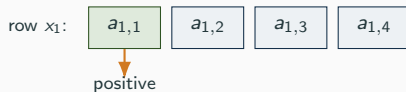
One Row of the Contrastive Matrix

row x_1 :

$a_{1,1}$	$a_{1,2}$	$a_{1,3}$	$a_{1,4}$
-----------	-----------	-----------	-----------

(1) Row i : one positive on the diagonal, $m - 1$ negatives.

One Row of the Contrastive Matrix



- (1) Row i : one positive on the diagonal, $m - 1$ negatives.
- (2) The positive logit is $a_{i,i} = u_i^\top v_i / \tau$.

One Row of the Contrastive Matrix



- (1) Row i : one positive on the diagonal, $m - 1$ negatives.
- (2) The positive logit is $a_{i,i} = u_i^\top v_i / \tau$.
- (3) Off-diagonal logits are in-batch negatives.

One Row of the Contrastive Matrix



row softmax over captions:

$$P_{1,j}^{I \rightarrow T} = \frac{\exp(a_{1,j})}{\sum_k \exp(a_{1,k})}$$

$$\text{loss} = -\log P_{1,1}^{I \rightarrow T}$$

- (1) Row i : one positive on the diagonal, $m - 1$ negatives.
- (2) The positive logit is $a_{i,i} = u_i^\top v_i / \tau$.
- (3) Off-diagonal logits are in-batch negatives.
- (4) Softmax cross-entropy turns the row into a loss.

Raising any negative's score hurts the loss even if the positive is unchanged.

Worked Contrastive Row

Caption logits for one image: (2.0, 0.0, 1.0), first caption matched.

$$\ell = -\log \frac{e^2}{e^2 + e^0 + e^1} \approx -\log(0.665) \approx 0.408.$$

If the third logit rises from 1.0 to 2.0, the denominator grows and matched probability falls.

Temperature: with $b_1 > b_2$,

$$p_1(\tau) = \frac{1}{1 + \exp(-(b_1 - b_2)/\tau)} = \text{sigmoid}\left(\frac{b_1 - b_2}{\tau}\right).$$

Smaller $\tau \Rightarrow$ sharper softmax over candidates.

Temperature controls how aggressively negatives compete with the positive.

Contrastive Gradient on the Image Embedding

Hold captions fixed. With $a_{i,j} = u_i^\top v_j / \tau$,

$$\nabla_{u_i} \ell_i^{I \rightarrow T} = \frac{1}{\tau} \left(\sum_{j=1}^m P_{i,j}^{I \rightarrow T} v_j - v_i \right).$$

Gradient descent subtracts this vector:

- pulls u_i *toward* the matched v_i ,
- pushes u_i *away from* captions with too much current probability.

Batch composition matters: false negatives or near-duplicate captions appear directly in the weighted negative average.

The denominator is why contrastive training is a ranking problem inside a batch.

Zero-Shot Classification

Class Names as Text Vectors

Prompt set for K labels: $c_1, \dots, c_K$, encoded to normalized vectors $v_1, \dots, v_K$. Nearest-text zero-shot classifier:

$$\hat{y}(x) = \arg \max_{1 \leq k \leq K} u(x)^\top v_k.$$

- No new classifier head trained for the named labels.
- Open-vocabulary: change the prompts, change the label set.
- Prompt scores are still over the *supplied* candidate set.

Zero-shot labels are text vectors evaluated at decision time.

Prompt Sensitivity and Ensembling

Same image, different prompt templates:

Template	Bicycle	Bus	Prediction
a photo of a {label}	0.62	0.59	bicycle
a campus scene with a {label}	0.55	0.64	bus
an object near a building: {label}	0.60	0.60	tie

Prompt-ensemble score:

$$s_k(x) = \frac{1}{R} \sum_{r=1}^R u(x)^\top v_{k,r}, \quad \hat{y}(x) = \arg \max_k s_k(x).$$

Prompts are part of the mathematical input; record and audit them like any other input.

Retrieval and Ranking

Ranks, Recall, MRR, Average Precision

Rank

Position of the matched item in the sorted candidate list, 1-based.

Recall@k

Fraction of queries whose first correct candidate has rank $\leq k$.

Mean reciprocal rank

$$\text{MRR} = \frac{1}{Q} \sum_{q=1}^Q \frac{1}{r_q}.$$

Average precision (one query)

$$\text{AP} = \frac{1}{R} \sum_{r=1}^M P@r \cdot \text{rel}(r), \quad P@r = \frac{1}{r} \sum_{j=1}^r \text{rel}(j).$$

Retrieval metrics evaluate *ordering*, not calibrated probability.

Ranked-List Diagnostics, Step by Step

query t

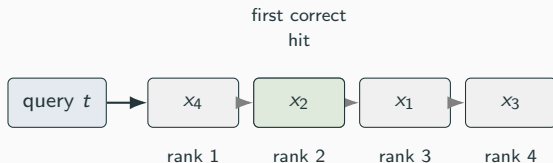
(1) Encode the query once.

Ranked-List Diagnostics, Step by Step



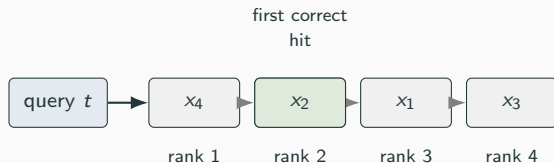
- (1) Encode the query once.
- (2) Sort candidates by similarity.

Ranked-List Diagnostics, Step by Step



- (1) Encode the query once.
- (2) Sort candidates by similarity.
- (3) Locate the first correct hit.

Ranked-List Diagnostics, Step by Step



ranks across queries: 1, 3, 2, 4
recall@2 = $2/4 = 0.5$
MRR = $\frac{1}{4}(1 + \frac{1}{3} + \frac{1}{2} + \frac{1}{4}) \approx 0.52$

- (1) Encode the query once.
- (2) Sort candidates by similarity.
- (3) Locate the first correct hit.
- (4) Aggregate ranks into recall@ k and MRR.

MRR rewards earlier first hits more aggressively than recall@10.

Average Precision Example

One image query with two correct captions at ranks 2 and 4 in a list of M candidates.

$$P@2 = \frac{1}{2}, \quad P@4 = \frac{2}{4}.$$

$$AP = \frac{1}{2} \left(\frac{1}{2} + \frac{2}{4} \right) = 0.50.$$

- Use AP when several captions or images are valid matches.
- Recall and AP ask different questions about the same list.
- Match the metric to the workflow: search, classify, calibrate.

A high retrieval score can still hide class-level errors; state which question the metric answers.

Fusion and Cross-Attention

Dual Encoder vs. Fusion Model

Dual encoder

Image vector $u = f_I(x)$ and text vector $v = f_T(t)$ are computed *separately*; score is $u^\top v$.

Fusion model

Representation depends on image *and* text tokens jointly *before* scoring; tokens interact.

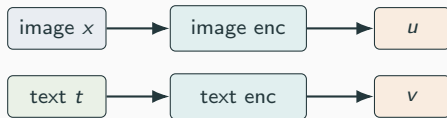
Cross-attention: text queries read from visual keys/values,

$$\text{softmax}_{\text{row}}\left(\frac{Q_T K_I^\top}{\sqrt{d_k}}\right) V_I.$$

Score matrix $T_T \times T_I$, cost $O(T_T T_I d_k)$ per head.

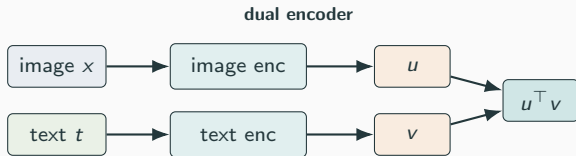
Dual encoders precompute; fusion models recompute per pair.

Architecture, Built Up



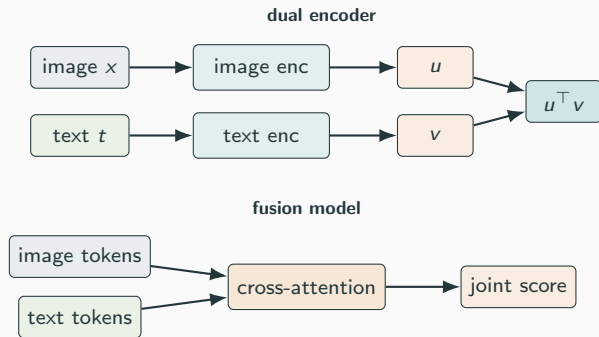
(1) Two encoders, two vectors.

Architecture, Built Up



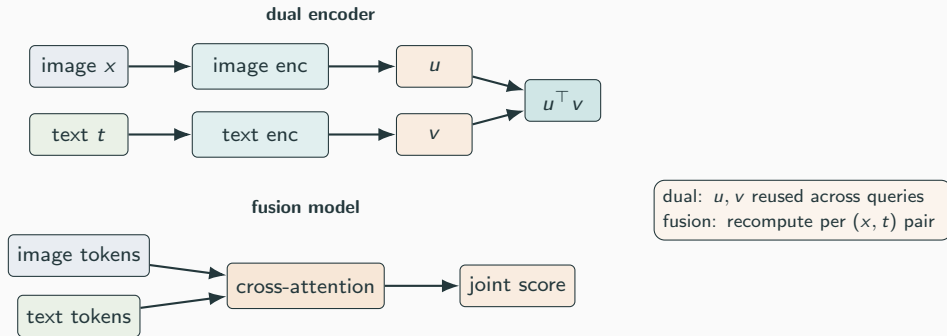
- (1) Two encoders, two vectors.
- (2) Dual scoring is a single dot product.

Architecture, Built Up



- (1) Two encoders, two vectors.
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- (3) Fusion lets tokens interact across modalities.

Architecture, Built Up



- (1) Two encoders, two vectors.
- (2) Dual scoring is a single dot product.
- (3) Fusion lets tokens interact across modalities.
- (4) Retrieval vs. grounding tradeoff is structural.

Pick the architecture to match query volume and required grounding detail.

Cross-Attention by Hand

Word red has scores $(2, 0, -1)$ over three image tokens. Row-softmax weights:

$$(0.844, 0.114, 0.042).$$

Scalar values $(1, 4, 10)$ give attended scalar

$$0.844(1) + 0.114(4) + 0.042(10) \approx 1.72.$$

- Word mostly reads from the first image token, but other tokens still contribute.
- Useful when the task needs phrase-region grounding (e.g. “on the chair” resolution).
- Memory per head: $T_T T_I$ entries.

Cross-attention is the same algebraic mechanism as sequence-to-sequence attention, with modalities swapped.

Diagnostics, Biases, Deployment

Things That Can Go Wrong

Shortcut correlation

A feature predictive in training data that is not the intended concept: background, watermark, camera angle, co-occurring object.

Modality gap

Systematic separation between image-side and text-side embeddings even after shared-space training.

Deployment shift

Deployment image-text distribution differs from training or validation distribution.

These are failures of the learned *geometry* under new conditions, not of matrix multiplication.

Failure-Mode Field Guide

Observed behavior	Likely issue	Applied check
Class flips after small prompt rewrite	prompt sensitivity	evaluate several prompt templates
Nearest images share background, not object	shortcut correlation	inspect nearest-neighbor grids
High similarity, low observed accuracy	miscalibration	calibrate or avoid probability claims
Good validation retrieval, poor field retrieval	deployment shift	test on deployment-like images
Correct object, wrong relation	weak token-level grounding	compare with a fusion model

Cosine similarity is a geometric score, not a calibrated probability of truth.

Inspecting the Neighborhood

Nearest-neighbor inspection usually exposes the problem first.

- Query “backpack on a chair” retrieves chairs without backpacks $\Rightarrow$ wrong visual evidence.
- Retrieves backpacks on floors, not chairs $\Rightarrow$ object captured, relation lost.
- Same sunny background across many queries $\Rightarrow$ background-as-shortcut.

Workflow checks:

- Sweep prompts; report sensitivity.
- Plot nearest-neighbor grids per class.
- Evaluate on deployment-like splits before release.

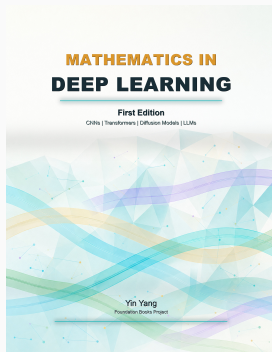
Inspect neighborhoods before trusting a shared embedding space.

Summary

Chapter in One Slide

- **Tokens:** an image becomes patches, patches become $\mathbb{R}^d$ tokens, plus class and position; $N = HW/P^2$.
- **ViT algebra:** self-attention has T^2 scores per head and $4d^2$ projection parameters per block.
- **Pooling:** class-token or average pooling produces one vector $r(x) \in \mathbb{R}^d$ for downstream heads.
- **Shared geometry:** normalize image and text vectors; cross-modal score is a cosine similarity.
- **Contrastive training:** the diagonal is the positive; off-diagonal entries are in-batch negatives.
- **Zero-shot:** text vectors of class prompts replace a learned classifier head.
- **Retrieval:** recall@k, MRR, AP evaluate ordering.
- **Fusion:** cross-attention adds token-level interaction at scoring cost.
- **Diagnostics:** prompts, neighborhoods, deployment splits.

Next: generative image models, diffusion, and structured latent spaces.



Mathematics in Deep Learning

Chapter overview slides for the book by Yin Yang.

Book page	foundation-books.github.io/mathematics-in-deep-learning
Code	github.com/foundation-books/mathematics-in-deep-learning-code
Paperback	amazon.com/dp/BOH1ZZHDT2
Hardcover	amazon.com/dp/BOH1ZL11MJ
Kindle	amazon.com/dp/B0GX32M8SP
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